

Research Methodology in CS (CSEG3060_3)

Lecture 1: Scope of Patent Rights - extent of protection, rights conferred, and limitations

Unit 6: Patent Rights and New Developments in IPR • Mapped Course Outcome: CO3

Research Methodology in Computer Science (CSEG3060_3)

Unit 6: Patent Rights and New Developments in IPR

Lecture 1: Scope of Patent Rights — Extent of Protection, Rights Conferred, and Limitations

1. Introduction

A patent is a statutory right granted by the sovereign state to an inventor, conferring upon the patentee the exclusive privilege to exclude third parties from commercially exploiting the invention for a limited period, in exchange for a full and public disclosure of the invention. While this statement captures the *essence* of a patent, it conceals an entire body of jurisprudence on the precise *scope* of the right — what it actually protects, what acts it prohibits, what the patentee is permitted to do, and what activities remain outside the patentee's control.

Understanding the **scope of patent rights** is foundational to three intersecting disciplines that a computer science researcher must master:

1. **Innovation strategy** — deciding what to patent and how broadly.
2. **Freedom-to-operate analysis** — ensuring that one's own software, algorithms, or systems do not fall within another's patent claims.
3. **Technology commercialisation** — valuing patents, licensing them, and negotiating technology transfer agreements.

This lecture systematically analyses the three pillars of patent scope: the **extent of protection** (subject matter and claim coverage), the **rights conferred** (positive and negative entitlements of the patentee), and the **limitations** (statutory and judicial carve-outs that restrain those rights). Throughout, the discussion is anchored to computer science and software-related innovations, since the subject is taught within a B.Tech (CS) programme and is examined under Course Outcome CO3.

2. Learning Objectives

By the end of this lecture, the student should be able to:

- Define the legal concept of "scope of patent rights" and articulate its territorial and temporal dimensions.
- Distinguish between the *subject matter* that may be patented and the *acts* that the patentee may control.
- Identify the statutory rights conferred by a patent and explain the doctrine of infringement.
- Enumerate the principal limitations on patent rights and apply them to software-related inventions.
- Critically analyse leading case law that has shaped the boundaries of patent protection in the computing domain.

3. Key Concepts and Terminology

Term	Definition
Patent	A time-limited, territorial statutory monopoly granted for a novel, non-obvious, and industrially applicable invention.
Scope of Patent Rights	The total bundle of exclusive privileges and prohibitions that flow from the grant, bounded by the <i>claims</i> and by statutory limitations.
Claims	The legally operative part of the patent specification that defines the metes and bounds of the invention.
Infringement	Any act that falls within the scope of the patented claims without the patentee's authorisation.
Exhaustion of Rights	The doctrine that the patentee's rights are extinguished upon a lawful first sale of a patented article.
Compulsory Licence	A non-voluntary licence granted by the State on prescribed grounds, allowing third parties to work the patent without the patentee's consent.
Prior User Right	The defence available to a person who was using the invention in secret before the patent's priority date.
Bolar Exception	A research exemption permitting generic manufacturers to use a patented invention for regulatory approval before patent expiry.

4. Extent of Protection

4.1 The Patent Specification and the Role of Claims

The *extent* of protection is determined principally by the **claims** of the patent. This is universally accepted: Section 10(5) of the Indian Patents Act, 1970 (as amended in 2005), Article 69 of the European Patent Convention, and 35 U.S.C. § 112 all converge on the principle that the claims define the scope. Everything outside the claims — the abstract, the description, the drawings — is context for interpretation but does not, by itself, confer protection.

In computing, claims are often drafted in multiple forms:

- **Apparatus/system claims** — directed to a hardware configuration, e.g., a multiprocessor architecture with a particular cache-coherence protocol.
- **Method/process claims** — directed to a sequence of steps, e.g., a method of compressing video using a specific predictive algorithm.
- **Computer-readable medium (CRM) claims** — directed to a non-transitory storage medium carrying instructions to perform the method; historically important in the United States.
- **Computer-implemented invention (CII) claims** — drafted as a hybrid often used before the European Patent Office.

A useful heuristic for the student: the broader the claim, the wider the scope of protection, but also the greater the risk of rejection over prior art or invalidation for lack of novelty or inventive step.

4.2 Subject Matter of Protection

Patent protection extends only to **patentable subject matter**. In computer science, this question has generated decades of litigation.

Under the Indian Patents Act, 1970 (Section 3): the following are expressly excluded from patentability:

- *Per se* computer programs (Section 3(k)) — although the Supreme Court, in *State of Street Bank and Trust Co.* (U.S.) and *Diamond v. Diehr* (U.S.), and the Indian Patents Office Manual (2011) recognise that a computer program combined with a novel hardware or technical effect is not "per se".
- Mathematical or business methods (Section 3(k)).
- Algorithms (Section 3(k)).
- Topographies of integrated circuits (Section 3(l)).
- Traditional knowledge (Section 3(p)).

Under U.S. law (35 U.S.C. § 101): subject matter is determined by the two-step *Alice/Mayo* framework. An abstract idea implemented on a generic computer is not patent-eligible. This led to the landmark decision in *Alice Corp. v. CLS Bank International* (2014), which invalidated software-related business method patents that claimed no inventive concept beyond the abstract idea.

Under the European Patent Convention (Article 52): programs for computers and mathematical methods are excluded "as such", but technical character can restore patentability (T 258/03 *Auction method*; T 1193/18 *G1/19*).

For the CS researcher, the practical takeaway is: **a pure algorithm, formula, or business method cannot be patented; but an algorithmic invention that produces a technical effect beyond the ordinary interaction between software and hardware may be.**

4.3 Territorial Extent

Patents are strictly **territorial** rights. A U.S. patent has no legal force in India, and an Indian patent is unenforceable in Germany. To obtain multi-jurisdictional protection, the inventor must file in each country of interest, ordinarily within twelve months of the priority date under the Paris Convention. This territoriality is fundamental to **freedom-to-operate (FTO)** analyses: a developer shipping software worldwide must examine patent registers in every relevant jurisdiction.

4.4 Temporal Extent (Term of Protection)

The standard term in most jurisdictions is **twenty years from the date of filing**, subject to the payment of renewal/annuity fees. India follows this norm under Section 53 of the Patents Act, 1970. Once the term expires, the patented subject matter enters the public domain, free for all to use. There is no concept of "evergreen" patents beyond this term, although supplementary protection certificates (SPCs) exist for pharmaceuticals in Europe.

5. Rights Conferred

5.1 The Negative Nature of the Right

A patent does not grant the patentee the *affirmative* right to make, use, or sell the invention. Rather, it grants the right to **prevent others** from doing so. This subtle but important distinction is fundamental to the operation of the patent system: the patent confers a "negative right" — it empowers exclusion, not performance.

In computing, this is illustrated starkly: a patent on a novel data-compression algorithm does not license the inventor to encode video; it only empowers the inventor to stop third parties from encoding video using that algorithm without consent.

5.2 The Statutory Bundle of Rights

Section 48 of the Indian Patents Act, 1970, sets out the rights conferred by a patent. The patentee has the exclusive right:

1. Where the invention is a **product**, to prevent third parties from making, using, offering for sale, selling, or importing that product for the purposes of commercial exploitation.
2. Where the invention is a **process**, to prevent third parties from using that process, and from using, offering for sale, selling, or importing the product obtained directly by that process.

This language is mirrored in Article 28 of the TRIPS Agreement, 1994, and is the universal standard.

5.3 Specific Acts of Infringement in Computing

The following acts, when undertaken without authorisation and within the scope of the claims, constitute infringement:

Act	Illustration in CS
Making	Manufacturing a chip embodying a patented architecture.
Using	Running a patented machine-learning inference method on a server.
Offering for sale	Advertising a software-as-a-service (SaaS) product that implements a patented process.
Selling	Distributing licensed copies of software that include a patented data structure.
Importing	Bringing into the territory hardware manufactured abroad that infringes a patented apparatus claim.

Indirect infringement (under Section 114A of the Indian Act and analogous provisions elsewhere) addresses contributory liability — supplying a means for working the invention when the supplier knows that the means are intended for such infringing use.

5.4 Rights Against Infringement

The patentee may enforce the right through:

- **Civil action** — injunction, damages or account of profits, delivery-up of infringing goods.
- **Administrative action** — seizure of infringing goods at customs (under Indian Customs Act, 1962, Section 11).

- **Criminal action** — though in most jurisdictions patent infringement is a civil wrong, not a crime; criminal sanctions for software piracy typically arise under copyright, not patent.
-

6. Limitations on Patent Rights

Limitations balance the patentee's exclusive right with public interest, enabling follow-on innovation and preventing abuse. They fall into two broad categories: **statutory exceptions** and **judicially evolved doctrines**.

6.1 Statutory Exceptions

6.1.1 Exhaustion of Rights (First Sale Doctrine) Once a patented product is sold lawfully by or with the consent of the patentee, the patentee's rights in respect of that specific article are exhausted. The buyer may use, resell, or repair the article freely. This is critical in computing: a customer who lawfully purchases a patented microprocessor may install it in any device without further licence from the patentee (parallel imports raise distinct questions under regional exhaustion).

6.1.2 Research and Experimental Use Exception Under Section 47(3) of the Indian Patents Act, acts done for experimental purposes relating to the subject matter of the invention do not constitute infringement. The scope is narrower in India than in some jurisdictions; in Europe, the experimental use defence is broad and covers research aimed at technical improvement. In the United States, the defence is limited by the decision in *Madey v. Duke University* (2002), which held that even non-profit university research with commercial implications falls outside the exception.

6.1.3 Bolar Exception Under Section 47(1)(d) and Explanation (a), acts reasonably related to obtaining regulatory approval for a product before patent expiry are exempted. This is the "Bolar exception", derived from the U.S. case *Roche Products v. Bolar Pharmaceutical* (1984), designed to prevent patent term de facto extension via regulatory lag. While the doctrine was originally formulated for drugs, the statutory language in India is not so limited.

6.1.4 Prior User Rights Section 47(2) of the Indian Patents Act protects a person who was in bona fide use of the invention in India before the patent's priority date. This is vital for the software industry, where trade-secret developers frequently claim that they were practising the algorithm or system privately. The Indian statute requires that the prior use must have been before the priority date and that the user must continue to use the invention.

6.1.5 Government Use Section 47(1)(a)–(c) and Chapter XVI of the Act permit the Central Government to use a patented invention for its own purposes on specified grounds. This is a form of *compulsory licence*.

6.1.6 Compulsory Licensing Section 84 of the Act empowers any interested party to apply for a compulsory licence three years after the patent's seal date, on grounds such as non-working, unreasonable refusal to licence, or public health necessity. The first Indian compulsory licence was granted in 2012 (*Natco v. Bayer, Nexavar*). While not yet a software matter, compulsory licensing remains doctrinally relevant to standard-essential patents (SEPs) in ICT.

6.2 Judicially Evolved Limitations

6.2.1 The Doctrine of Equivalents Developed in the United States (and applied cautiously in India), the doctrine prevents an infringer from escaping liability by making insubstantial modifications to a patented invention. However, it has been substantially narrowed by *Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co.* (2002), which imposed prosecution-history estoppel on the doctrine's application. In CS, this often arises when a competitor writes the algorithm in a slightly different programming language or uses a different hardware platform.

6.2.2 Inequitable Conduct and Patent Misuse A patentee who withholds material prior art from the examiner, or who extends the scope of the patent beyond its legitimate boundaries (e.g., by tying sales of patented software to other products), may be barred from enforcement.

6.2.3 Abuse of Dominance In competition law, particularly Article 102 of the Treaty on the Functioning of the European Union and Sections 4 of the Indian Competition Act, 2002, a dominant patent holder's refusal to license standard-essential patents may constitute an abuse. The *Huawei v. ZTE* (2015) and *Microsoft v. Motorola* (2013) line of cases illustrates the intersection of FRAND commitments with patent scope.

7. Case Studies

7.1 *Diamond v. Diehr*, 450 U.S. 175 (1981)

Facts: Diehr claimed a process for curing rubber that used a computer to calculate cure times based on the Arrhenius equation.

Holding: The U.S. Supreme Court held that a claim that recites a mathematical formula does not make it unpatentable so long as the formula is applied in a non-abstract, practical manner producing a transformation.

Significance for CS: Established that software-implemented processes with a tangible technical effect are patent-eligible. Remains a foundational citation for software patents worldwide.

7.2 *Alice Corp. v. CLS Bank International*, 573 U.S. 208 (2014)

Facts: Alice claimed a computer-implemented method for mitigating settlement risk using a third-party intermediary.

Holding: Applying the two-step *Alice/Mayo* test, the Supreme Court held the claims invalid as directed to an abstract idea (intermediated settlement) without an inventive concept.

Significance: Effectively narrowed software patent eligibility, particularly for business methods. Has been used to invalidate hundreds of subsequent software claims in the United States.

7.3 *State of Street Bank and Trust Co. v. Signature Financial Group* (1998, U.S. CAFC)

Facts: Signature patented a data processing system for managing mutual fund settlements.

Holding: The Court held that business methods implemented on a computer are patentable so long as they produce a "useful, concrete, and tangible result".

Significance: Opened the floodgates to U.S. business method patents in the late 1990s. Subsequently limited by *Bilski* (2010) and *Alice* (2014).

7.4 *Novartis AG v. Union of India*, AIR 2013 SC 1316 (Evergreening)

While primarily a pharmaceutical case, this Supreme Court decision is instructive because it applied Section 3(d) to deny a patent on the basis of *evergreening* — minor variations of known substances not showing enhanced efficacy. The reasoning has been cited by CS commentators to argue that incremental software updates that do not contribute technical effect should similarly fail the inventive-step test.

7.5 *Koninklijke Philips Electronics NV v. Princo Corporation* (Italy/EPO, 2009)

A leading European case on the exhaustion doctrine applied to CD-R patents. Demonstrates that even where a patent has been exhausted, related patents may still be enforceable where the exhaustion argument fails for different claim categories. Relevant to layered software-hardware patent portfolios.

8. Practical Implications for Computer Scientists

1. **Drafting Claims Carefully** — A researcher filing a patent on a novel algorithm must draft at least one claim directed to a technical application (e.g., a method of video encoding implemented in hardware) to avoid rejection under Section 3(k).

2. **Freedom-to-Operate (FTO) Analyses** — Before launching a new SaaS product or releasing a library, a startup should conduct FTO in major markets. A single unaddressed patent in the portfolio of a non-practising entity (NPE, colloquially "patent troll") can result in injunctions or settlements.
 3. **Open Source Compatibility** — Many open source licences (GPL, LGPL, Apache) interact with patents differently. The Open Patent Non-Aggression Pledge (now OIN) is a defensive patent pool relevant to Linux and adjacent systems.
 4. **Standard-Essential Patents (SEPs)** — If the researcher's work contributes to a standard (e.g., 5G, Wi-Fi, MPEG), licensing obligations under FRAND terms may be imposed, restricting the scope of rights that can be enforced.
 5. **Trade Secrets vs. Patents** — Algorithms that can be reverse-engineered from products are often better protected by patents. Algorithms that derive value from secrecy (e.g., proprietary machine-learning model weights) are better protected by trade secret.
-

9. Summary

- The **scope of patent rights** is determined by the claims, bounded by statutory subject-matter limits, and confined to the territory of grant.
 - Patents confer a **negative right** to exclude third parties from making, using, offering for sale, selling, or importing the patented invention.
 - The **extent of protection** is narrower than the description; broad claims carry higher invalidation risk.
 - **Limitations** include exhaustion of rights, research and experimental use, the Bolar exception, prior user rights, government use, compulsory licensing, and judicially evolved doctrines such as equivalents and inequitable conduct.
 - In computing, the boundary between patentable and non-patentable subject matter is contested but generally requires a **technical effect** beyond *per se* algorithms or business methods.
 - Case law — particularly *Diamond v. Diehr* and *Alice v. CLS Bank* — governs the contemporary trajectory of software patentability.
-

10. Practice and Exam-Oriented Questions

Short-Answer Questions (2–3 marks)

1. Distinguish between the *negative* and *positive* nature of patent rights. Why is this distinction important in software commercialisation?
2. Define *exhaustion of rights*. How does it apply when a software product is lawfully sold under a licence?
3. What is the *Bolar exception*, and why is it relevant to the CS research community?
4. Briefly explain the territorial nature of patents and its consequence for freedom-to-operate analyses.
5. State the role of *claims* in determining the scope of patent protection.

Long-Answer Questions (10–15 marks)

1. Critically analyse the scope of patent protection for computer-implemented inventions under Indian law. Compare and contrast with the position in the United States after *Alice Corp. v. CLS Bank*.
2. Discuss the various statutory limitations on patent rights. Using the prior user right, evaluate its adequacy for the Indian software industry.
3. Examine the rights conferred under Section 48 of the Indian Patents Act, 1970. How are these rights enforced against indirect infringement in software supply chains?
4. "The scope of patent rights is a balance between innovation incentives and public interest." Discuss with reference to *Diamond v. Diehr*, *Alice v. CLS Bank*, and *Novartis v. Union of India*.
5. Compare the doctrines of exhaustion of rights, compulsory licensing, and government use as limitations on patent rights. Which is most relevant to standard-essential patents in 5G?

Higher-Order / Application Questions

1. A small AI startup in Bengaluru develops a novel training algorithm for large language models. Advise the founders on: (a) whether the algorithm itself is patent-eligible in India, the U.S., and the EU; (b) whether publishing a research paper before filing would destroy novelty; (c) how to conduct a basic FTO before commercial launch.
2. A non-practising entity sends a cease-and-desist letter to your employer alleging that an open-source middleware component infringes its U.S. patent. Draft a reasoned response addressing: scope of the U.S. patent, exhaustion, and experimental use defences.

11. Suggested Reading

1. **Intellectual Property Rights in the Knowledge Economy** — IP Raj, LexisNexis (latest edition).
2. **Indian Patents Act, 1970**, with commentary by N.S. Gopalakrishnan and T.G. Agitha.
3. **Patent Law** (4th ed.) — Robert P. Merges and John F. Duffy, Wolters Kluwer.

4. **Software Patents: A Practical Guide** — Vernon W. French, Oxford University Press.
 5. **The Alice/Mayo Two-Step Framework**, 35 U.S.C. § 101 — official USPTO guidance (rev. ed.).
 6. **Manual of Patent Office Practice and Procedure** (India), Office of the Controller General of Patents, Designs & Trade Marks.
 7. WIPO Lex — International Patent Law Database: <https://wipolex.wipo.int>.
-

End of Lecture 1 Notes. These notes map directly to Course Outcome CO3 of CSEG3060_3, equipping students to critically evaluate the scope, rights, and limitations of patents in computing research and innovation contexts.